MTH210

Examination - January Semester 2025

Fundamentals of Probability

Thursday, 17 April 2025 **04:00 pm - 06:00 pm**

Time allowed: 2 hours

INSTRUCTIONS TO STUDENTS:

1. This examination contains **FOUR (4)** questions and comprises **FOUR (4)** printed pages (including cover page).

2. You must answer **ALL** questions.

3. All answers must be written in the answer book. Marks will only be awarded if FULL working is shown.

4. This is an **Open-book** examination.

At the end of the examination.
Please ensure that you have written your examination number on each answer book used.

Failure to do so will mean that your work cannot be identified.
If you have used more than one answer book, please tie them together with the string provided.

THE UNIVERSITY RESERVES THE RIGHT NOT TO MARK YOUR SCRIPT IF YOU FAIL TO FOLLOW THESE INSTRUCTIONS.

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Answer all questions. (Total: 100 marks)

Question 1

Given that the sample space is $S = \{5, 6, 7, 8, 9, 10, 11, 12\}$. Let $E_1 = \{6, 8, 10\}$ and $E_2 = \{8, 9, 10, 11\}$.

(a) Calculate the probabilities $P(E_1 \cup E_2)$ and $P(E_1 \cap E_2)$.

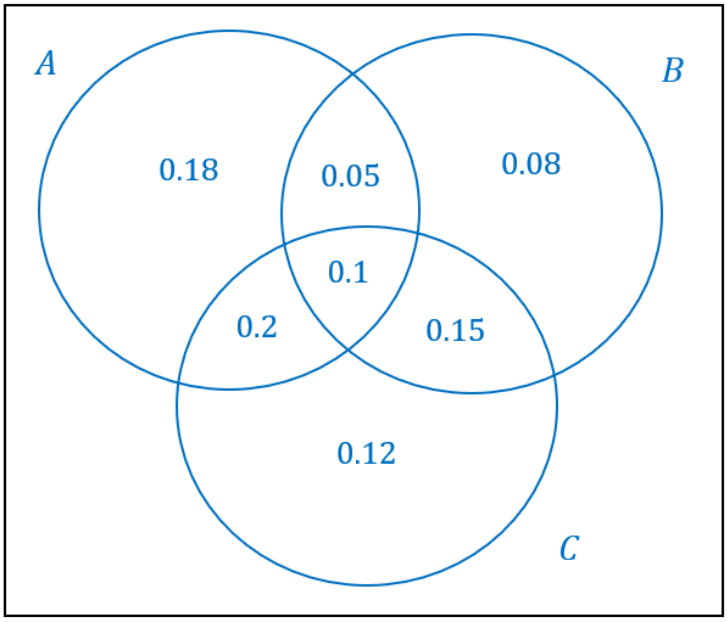
(12 marks)

(b) Calculate the probabilities $P((E_1 \cap E_2)')$ and $P((E_1' \cap E_2)')$.

(13 marks)

Question 2

(a) A study was conducted to find out about the reading habits of readers. The readers were asked what types of books they read. There are three choices - "science fiction" (A), "personal improvement" (B) and "design book" (C). The Venn diagram below shows the probabilities of the different events.



(i) Compute the probability that a randomly selected reader reads "science fiction", given that he reads at least one type of books.

(6 marks)

(ii) Compute $P(A \cup B|C)$.

(6 marks)

(b) Given that

$$P(A) = 0.75, P(B|A) = 0.9, P(B|A') = 0.8,$$

compute the following probabilities with the aid of a Venn diagram or otherwise.

(i) $P(A \cap B)$

(6 marks)

(ii) $P(B)$

(7 marks)

Question 3

(a) The probability mass function (pmf) for a discrete random variable Y is given below.

y	1	2	4	8	16
$P(Y = y)$	0.05	0.10	a	0.40	0.10

(i) Compute the probability of obtaining either a 1 or 16, i.e. $P(Y = 1$ or $16)$.

(5 marks)

(ii) Using the value of a that you have found in Question 3a(i), solve the cumulative distribution for Y in the last row of the table.

y	1	2	4	8	16
$P(Y = y)$	0.05	0.10	a	0.40	0.10
$F(y)$					

(5 marks)

(b) The probability density function (pdf) of a continuous random variable X is given by:

$$f(x) = \begin{cases} k[1-(x-3)^2], & 2 \le x \le 4 \\ 0, & \text{otherwise} \end{cases}$$

where k is an unknown constant.

(i) Determine the value of k.

(5 marks)

(ii) Determine the mean and variance of X.

(10 marks)

Question 4

(a) Defects occur in the sensors used for washing machine according to a Poisson model with a rate of 0.02 defect per sensor.

(i) Determine the probability that there is no defect in a randomly selected sensor.

(5 marks)

(ii) If 50 sensors are inspected, apply a suitable model and compute the probability that the number of sensors that have one or more defects is less than or equal to 2?

(7 marks)

(b) A company manufacture fish balls and squid balls and deliver them to supermarkets in boxes. The masses of the boxes of fish balls (in grams) follow normal distribution with mean 3400 and variance 350^2. The masses of the boxes of squid balls (in grams) also follow normal distribution with mean μ and variance σ^2.

(i) Compute the probability that a randomly selected box of fish balls has mass below 3200 grams.

(5 marks)

(ii) Let T be the total mass of 5 randomly selected boxes of fish balls and 3 randomly selected boxes of squid balls. Given that

$$P(T < 29084.5) = P(T > 31915.5) = 0.1.$$

Determine the value of μ and σ.

(8 marks)

----- END OF PAPER -----